**Relational Databases**

**1. What is normalisation?**

Normalisation is a technique used to organise a database efficiently. The goal is to minimise data redundancy and reduce the potential for data anomalies. It involves breaking down a database into smaller, related tables and defining relationships between them. This process improves data consistency and integrity.

**2. When is a table in 1NF?**

A table is in the First Normal Form (1NF) if:

● Each cell contains only one value.

● Each row is unique and identifiable by a primary key.

**3. When is a table in 2NF?**

A table is in the Second Normal Form (2NF) if:

● It meets all the rules of 1NF.

● All non-key attributes are fully dependent on the entire primary key. This means no column depends only on a part of a composite primary key.

**4. When is a table in 3NF?**

A table is in the Third Normal Form (3NF) if:

● It meets all the rules of 2NF.

● All non-key attributes depend only on the primary key and not on other non-key attributes. This removes transitive dependencies.

**5. INVOICE dependency diagram**

The relational scheme for the INVOICE table structure is as shown below:

INVOICE (INV_NUM, PROD_NUM, SALE_DATE, PROD_LABEL, VEND_CODE, VEND_NAME, QUANT_SOLD, PROD_PRICE)

The primary key for the table INVOICE is {INV_NUM, PROD_NUM}.

The functional dependencies are as follows: 1NF

{INV_NUM, PROD_NUM} {SALE_DATE, PROD_LABEL, VEND_CODE, VEND_NAME, QUANT_SOLD, PROD_PRICE}

{INV_NUM} {SALE_DATE}

{PROD_NUM} {PROD_LABEL, PROD_PRICE, VEND_CODE, VEND_NAME}

{VEN_CODE}{VEN_NAME}

Among the dependencies, the partial dependencies are as follows:

{INV_NUM}{SALE_DATE}

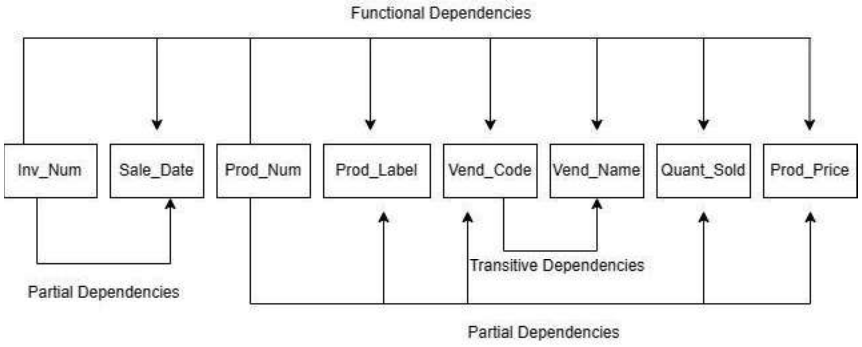
{PROD_NUM}{ PROD_LABEL, PROD_PRICE, VEND_CODE, VEND_NAME}

Among the dependencies, the transitive dependencies are as follows:

{VEN_CODE}{VEN_NAME}

The dependency diagram is as follows:

**Invoice Table Dependency Diagram**




**6. Remove all partial dependencies and draw the new dependency diagrams**

The functional dependencies are as follows: 2NF

{INV_NUM, PROD_NUM}{ QUANT_SOLD}

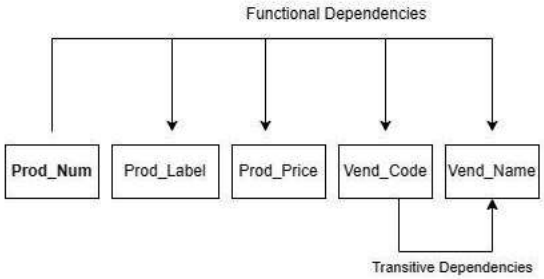
{INV_NUM}{SALE_DATE}

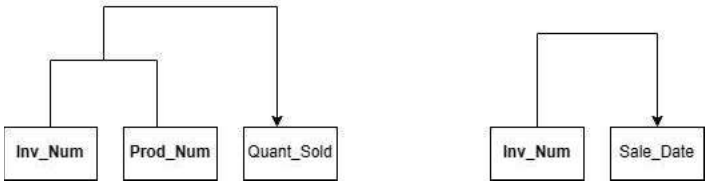
{PROD_NUM}{ PROD_LABEL, PROD_PRICE, VEND_CODE, VEND_NAME}

Among the dependencies, the transitive dependencies are as follows:

{VEN_CODE}{VEN_NAME}

**Invoice Table Dependency Diagram**




**7. Remove all transitive dependencies and draw the new dependency diagrams**

The functional dependencies are as follows: 3NF

{INV_NUM, PROD_NUM} { QUANT_SOLD}

{INV_NUM} {SALE_DATE}

{PROD_NUM} { PROD_LABEL, PROD_PRICE, VEND_CODE}

{VEN_CODE} {VEN_NAME}

## Invoice Table Dependency Diagram

